GECG10069 (561084) F26: Introduction to Programming (C++)

Homework 1 : Uniform-Invoice Lottery

HOMEWORK 1: UNIFORM-INVOICE LOTTERY

- ✓ In this homework, you are given an arbitrary number of uniform invoices (統一發票). Your task is to write a C++ program that can help check each invoice using the winning numbers of the 115 (2026) July–August Uniform Invoice Lottery, determine its highest prize, and calculate the total prize amount.
- ✓ In Taiwan, each Uniform Invoice contains a **two-letter invoice** track followed by an **8-digit invoice number**.
  - ➢ For example: `AB-12345678`, where AB is the invoice track and 12345678 is the invoice number.
- ✓ The winning numbers used in this problem are fixed as follows:

115 (2026) July–August Uniform Invoice Lottery

Prize	Winning Number / Matching Rule	Prize Amount
Special Prize	`89996565` (all 8 digits match)	NT$10,000,000
Grand Prize	`91098182` (all 8 digits match)	NT$2,000,000
First Prize	`54348835`, `44991397`, `06595111` (all 8 digits match any one)	NT$200,000
Second Prize	Last 7 digits match any First Prize number	NT$40,000
Third Prize	Last 6 digits match any First Prize number	NT$10,000
Fourth Prize	Last 5 digits match any First Prize number	NT$4,000
Fifth Prize	Last 4 digits match any First Prize number	NT$1,000
Sixth Prize	Last 3 digits match any First Prize number	NT$200

  - ➢ For the **Special Prize** and **Grand Prize**, all 8 digits must match the corresponding winning number.
  - ➢ For the **First Prize**, all 8 digits must match one of the three First Prize winning numbers.
  - ➢ For the **Second to Sixth Prizes**, only the corresponding ending digits of the 8-digit invoice number are compared with the First Prize winning numbers.
- ✓ If an invoice satisfies the conditions of more than one prize, **only the highest prize should be awarded**.
- ✓ Cloud Invoice Exclusive Prizes are not considered in this homework.

Input

- ✓ The input contains an arbitrary number of invoice records.
- ✓ Each invoice is given on a **separate line** in the following format: `AA-NNNNNNNN`

- ➢ `AA` consists of **exactly two uppercase English letters**.
- ➢ `NNNNNNNN` consists of **exactly eight digits from 0 to 9**.
  - ◆ The invoice number may contain leading zeros.

✓ The last line contains: `END`, which indicates the end of the input.

✓ There is **at least one invoice** before `END`.

Output

✓ For each invoice, output one line in the following format:

- ➢ <invoice>: <result>

✓ After processing all invoices, output the total prize amount:

- ➢ Total Prize: <amount>
- ➢ The total prize amount **will not exceed $2^{31} - 1$**
- ➢ Output the prize amount as an integer without commas.

✓ For NYCU students: For this homework, you must strictly follow the specified output format. Do not print any additional prompts, instructions, or unnecessary messages, as they may interfere with the automated grading script and result in incorrect evaluation.

Submission (For NYCU students)

✓ Please submit your solution code to Homework 1 to NTU COOL using the following filename format:

- ➢ [student_id]_hw1.cpp
- ➢ Example: 315511000_hw1.cpp

✓ Please strictly follow the required submission format and filename convention. If an incorrect filename or submission format prevents our automated grading script from compiling, executing, or evaluating your program correctly, you will be required to complete a make-up demo, and your score will be calculated according to the make-up submission grading policy.

Sample Input 1

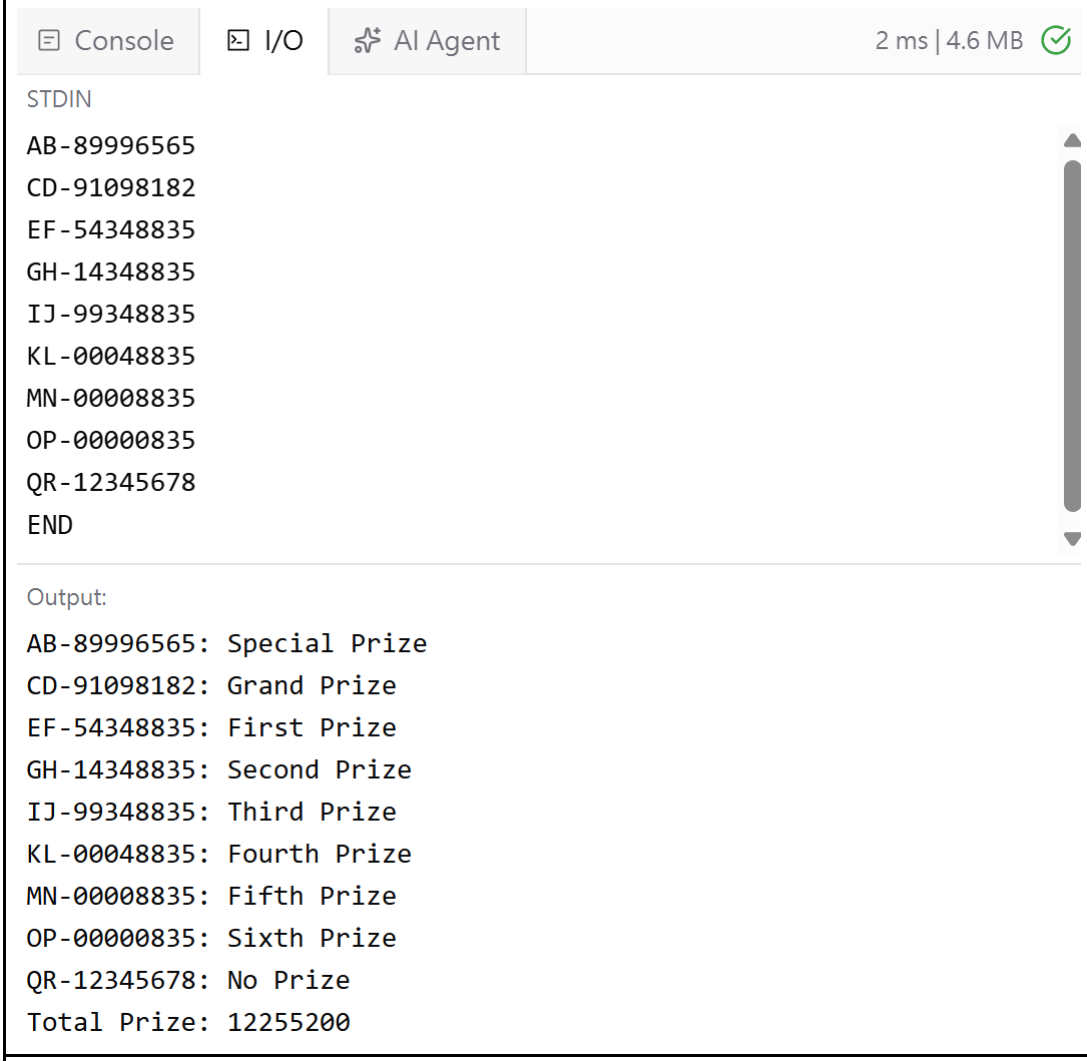
```
AB-89996565
CD-91098182
EF-54348835
GH-14348835
IJ-99348835
KL-00048835
MN-00008835
```

```
OP-00000835
QR-12345678
END
```

Sample Output 1

```
AB-89996565: Special Prize
CD-91098182: Grand Prize
EF-54348835: First Prize
GH-14348835: Second Prize
IJ-99348835: Third Prize
KL-00048835: Fourth Prize
MN-00008835: Fifth Prize
OP-00000835: Sixth Prize
QR-12345678: No Prize
Total Prize: 12255200
```

Sample Screenshot 1



Sample Input 2

```
AA-12345678
BC-87654321
DE-31415926
FG-27182818
HI-55555555
END
```

Sample Output 2

```
AA-12345678: No Prize
BC-87654321: No Prize
DE-31415926: No Prize
FG-27182818: No Prize
HI-55555555: No Prize
Total Prize: 0
```

Sample Screenshot 2

